Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Tony Borba Dairy Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
23844 E. Dodds RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 11/01/1955

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X207-X170-XX02-XXXX

B. OPERATORS

Borba, Tony

Operator name: Borba, Tony

Telephone no.: (209) 838-7018 Landline, (209) 604-1764 Cellular

Mailing Address Number and Street	City	State	Zip Code
23000 E Dodds RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Borba, Delores

Legal owner name: Borba, Delores

Telephone no.: (209) 838-7271 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
23900 E Dodds RD	Escalon	CA	95320

Borba, Tony

Legal owner name:	Borba, Tony	Telephone no.:	(209) 838-7018	(209) 604-1764
			Landline	Cellular
23000 E Dodds RD	Escalon		CA	95320
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	315	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	315	0	0	0
Average number	0	0	315	0	0	0
Avg live weight (lbs)	0	0	950	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 3,241.13 *tons per reporting period*

Total nitrogen from manure: 29,893.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 20,925.45 *lbs per reporting period*

Total phosphorus from manure: 5,058.90 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 6,840,000 *gallons*

Total nitrogen generated: 11,973.84 *lbs*

Total phosphorus generated: 2,000.87 *lbs*

Total potassium generated: 23,601.75 *lbs*

Total salt generated: 170,923.96 *lbs*

	6,840,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	6,840,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW F1	Ground water
IW Milk Barn	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Corral solids *Manure Import (Machado Dairy #4)*	1,000.00 *ton*	Dry-weight	64.5	25,100.00	19,400.00	22,000.00		46.00
05/07/2023	Corral solids *Manure Import (Machado Dairy #4)*	1,200.00 *ton*	Dry-weight	46.8	17,000.00	8,200.00	68,700.00		32.20

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
03/04/2023	Solid commercial fertilizer *CAN17*	8.55 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000
05/16/2023	Solid commercial fertilizer *UN32*	5.32 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
06/15/2023	Solid commercial fertilizer *4-0-0*	15.90 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	7,576.22	0.00	0.00	0.00
Dry manure	39,526.60	24,243.76	103,336.16	737,729.60
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	47,102.82	24,243.76	103,336.16	737,729.60

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	38	38	1	none	X207-X170-XX02-XXXX X207-X180-XX01-XXXX
Field 2	53	53	2	both	X207-X170-XX02-XXXX X207-X180-XX01-XXXX
Field 3	53	53	2	both	X207-X170-XX02-XXXX
Field 4	53	53	2	manure	X207-X170-XX02-XXXX
Totals for areas that were used for application	159	159	6		
Totals for areas that were not used for application	38	38	1		
Land application area totals	197	197	7		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/15/2017: Almond, in shell

Crop: Almond, in shell Acres planted: 38 Plant date: 10/15/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	50.10 *ton*	As-is		10.0	65,000.00	11,000.00	70,500.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.50	64.50	10.95	69.75	0.00
Total actual harvest content	1.32	171.39	29.01	185.90	0.00

Field 2

Field name: Field 2

Field 2

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	816.20 *ton*	As-is		61.0	3,280.00	1,010.00	13,700.00		6.73

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.40	101.02	31.11	421.96	808.41

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,462.80 *ton*	As-is		70.0	4,000.00	750.00	3,300.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.60	220.80	41.40	182.16	993.60

Field 3

Field name: Field 3

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	967.60 *ton*	As-is		73.0	2,190.00	590.00	9,320.00		9.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.26	79.96	21.54	340.30	898.12

Field 3

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,452.20 *ton*	As-is		70.0	4,000.00	750.00	3,300.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.40	219.20	41.10	180.84	986.40

Field 4

Field name: Field 4

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	853.30 *ton*	As-is		54.8	4,020.00	1,040.00	12,600.00		7.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.10	129.44	33.49	405.72	1,146.89

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,425.70 *ton*	As-is		70.0	4,400.00	750.00	3,300.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.90	236.72	40.35	177.54	968.40

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/15/2017: Almond, in shell

Field name: Field 1

Crop: Almond, in shell

Plant date: 10/15/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.57	0.00	0.00	160.38	3,843,700.00 *gal*
Application event totals		2.57	0.00	0.00	160.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.57	0.00	0.00	160.38	3,843,700.00 *gal*
Application event totals		2.57	0.00	0.00	160.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	38.25	0.00	0.00	0.00	
Application event totals		38.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	38.25	0.00	0.00	0.00	
Application event totals		38.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	44.80	0.00	0.00	0.00	
Application event totals		44.80	0.00	0.00	0.00	

Field 1 - 10/15/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.51	0.00	0.00	156.18	3,743,000.00 *gal*
Application event totals		2.51	0.00	0.00	156.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.51	0.00	0.00	156.18	3,743,000.00 *gal*
Application event totals		2.51	0.00	0.00	156.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	44.80	0.00	0.00	0.00	
Application event totals		44.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.51	0.00	0.00	156.18	3,743,000.00 *gal*
Application event totals		2.51	0.00	0.00	156.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.51	0.00	0.00	156.18	3,743,000.00 *gal*
Application event totals		2.51	0.00	0.00	156.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.58	0.00	0.00	160.54	3,847,500.00 *gal*
Application event totals		2.58	0.00	0.00	160.54	

Field 1 - 10/15/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.58	0.00	0.00	160.54	3,847,500.00 *gal*
Application event totals		2.58	0.00	0.00	160.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.58	0.00	0.00	160.54	3,847,500.00 *gal*
Application event totals		2.58	0.00	0.00	160.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.58	0.00	0.00	160.54	3,847,500.00 *gal*
Application event totals		2.58	0.00	0.00	160.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.58	0.00	0.00	160.54	3,847,500.00 *gal*
Application event totals		2.58	0.00	0.00	160.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	2.58	0.00	0.00	160.54	3,847,500.00 *gal*
Application event totals		2.58	0.00	0.00	160.54	

Field 2 - 10/31/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	168.12	129.94	147.36	3,081.13	500.00 *ton*
Application event totals		168.12	129.94	147.36	3,081.13	

Field 2 - 06/11/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	162.40	48.12	244.88	5,748.35	450.00 *ton*
Application event totals		162.40	48.12	244.88	5,748.35	

06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.90	0.00	0.00	118.47	3,960,000.00 *gal*
IW Milk Barn	Ground water	1.55	0.00	0.00	98.72	3,300,000.00 *gal*
Application event totals		3.45	0.00	0.00	217.19	

Field 2 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.70	11.12	202.74	1,362.38	1,710,000.00 *gal*
IW F1	Ground water	1.31	0.00	0.00	81.85	2,736,000.00 *gal*
IW Milk Barn	Ground water	1.07	0.00	0.00	68.21	2,280,000.00 *gal*
Application event totals		110.08	11.12	202.74	1,512.44	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.83	0.00	0.00	114.16	3,816,000.00 *gal*
IW Milk Barn	Ground water	1.49	0.00	0.00	95.13	3,180,000.00 *gal*
Application event totals		3.32	0.00	0.00	209.29	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.80	0.00	0.00	112.01	3,744,000.00 *gal*
IW Milk Barn	Ground water	1.46	0.00	0.00	93.34	3,120,000.00 *gal*
Application event totals		3.26	0.00	0.00	205.34	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.80	0.00	0.00	112.01	3,744,000.00 *gal*
IW Milk Barn	Ground water	1.46	0.00	0.00	93.34	3,120,000.00 *gal*
Application event totals		3.26	0.00	0.00	205.34	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.76	0.00	0.00	109.85	3,672,000.00 *gal*
IW Milk Barn	Ground water	1.44	0.00	0.00	91.54	3,060,000.00 *gal*
Application event totals		3.20	0.00	0.00	201.39	

Field 2 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.66	0.00	0.00	103.39	3,456,000.00 *gal*
IW Milk Barn	Ground water	1.35	0.00	0.00	86.16	2,880,000.00 *gal*
Application event totals		3.01	0.00	0.00	189.55	

Field 3 - 10/31/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.43	10.61	69.38	584.97	2,160,000.00 *gal*
Application event totals		31.43	10.61	69.38	584.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.12	10.17	66.49	560.60	2,070,000.00 *gal*
Application event totals		30.12	10.17	66.49	560.60	

Field 3 - 06/13/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	204.77	98.77	827.51	3,878.58	600.00 *ton*
Application event totals		204.77	98.77	827.51	3,878.58	

06/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.90	0.00	0.00	118.47	3,960,000.00 *gal*
IW Milk Barn	Ground water	1.55	0.00	0.00	98.72	3,300,000.00 *gal*
Application event totals		3.45	0.00	0.00	217.19	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.80	0.00	0.00	112.01	3,744,000.00 *gal*
IW Milk Barn	Ground water	1.46	0.00	0.00	93.34	3,120,000.00 *gal*
Application event totals		3.26	0.00	0.00	205.34	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.80	0.00	0.00	112.01	3,744,000.00 *gal*
IW Milk Barn	Ground water	1.46	0.00	0.00	93.34	3,120,000.00 *gal*
Application event totals		3.26	0.00	0.00	205.34	

Field 3 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.76	0.00	0.00	109.85	3,672,000.00 *gal*
IW Milk Barn	Ground water	1.44	0.00	0.00	91.54	3,060,000.00 *gal*
Application event totals		3.20	0.00	0.00	201.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.76	0.00	0.00	109.85	3,672,000.00 *gal*
IW Milk Barn	Ground water	1.44	0.00	0.00	91.54	3,060,000.00 *gal*
Application event totals		3.20	0.00	0.00	201.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.73	0.00	0.00	107.70	3,600,000.00 *gal*
IW Milk Barn	Ground water	1.41	0.00	0.00	89.75	3,000,000.00 *gal*
Application event totals		3.14	0.00	0.00	197.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.68	5.85	106.71	717.04	900,000.00 *gal*
IW F1	Ground water	1.73	0.00	0.00	107.70	3,600,000.00 *gal*
IW Milk Barn	Ground water	1.41	0.00	0.00	89.75	3,000,000.00 *gal*
Application event totals		59.82	5.85	106.71	914.49	

Field 4 - 11/01/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	168.12	129.94	147.36	3,081.13	500.00 *ton*
Application event totals		168.12	129.94	147.36	3,081.13	

Field 4 - 06/15/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	204.77	98.77	827.51	3,878.58	600.00 *ton*
Application event totals		204.77	98.77	827.51	3,878.58	

06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.94	0.00	0.00	120.62	4,032,000.00 *gal*
IW Milk Barn	Ground water	1.58	0.00	0.00	100.52	3,360,000.00 *gal*
Application event totals		3.51	0.00	0.00	221.14	

Field 4 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.83	0.00	0.00	114.16	3,816,000.00 *gal*
IW Milk Barn	Ground water	1.49	0.00	0.00	95.13	3,180,000.00 *gal*
Application event totals		3.32	0.00	0.00	209.29	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.80	0.00	0.00	112.01	3,744,000.00 *gal*
IW Milk Barn	Ground water	1.46	0.00	0.00	93.34	3,120,000.00 *gal*
Application event totals		3.26	0.00	0.00	205.34	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.80	0.00	0.00	112.01	3,744,000.00 *gal*
IW Milk Barn	Ground water	1.46	0.00	0.00	93.34	3,120,000.00 *gal*
Application event totals		3.26	0.00	0.00	205.34	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.73	0.00	0.00	107.70	3,600,000.00 *gal*
IW Milk Barn	Ground water	1.41	0.00	0.00	89.75	3,000,000.00 *gal*
Application event totals		3.14	0.00	0.00	197.45	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.73	0.00	0.00	107.70	3,600,000.00 *gal*
IW Milk Barn	Ground water	1.41	0.00	0.00	89.75	3,000,000.00 *gal*
Application event totals		3.14	0.00	0.00	197.45	

Field 4 - 06/15/2023: Corn, silage

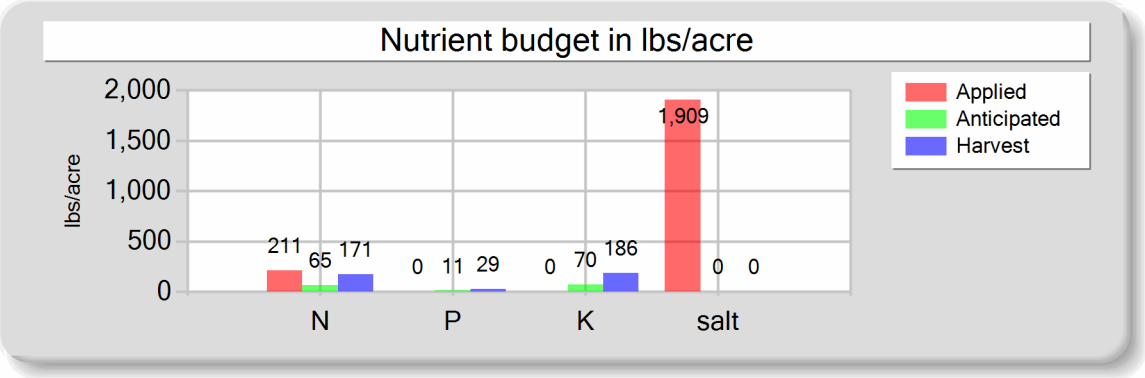
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW F1	Ground water	1.69	0.00	0.00	105.54	3,528,000.00 *gal*
IW Milk Barn	Ground water	1.38	0.00	0.00	87.95	2,940,000.00 *gal*
Application event totals		3.07	0.00	0.00	193.50	

B. NUTRIENT BUDGET

Field 1 - 10/15/2017: Almond, in shell

Field name: Field 1 Crop: Almond, in shell Plant date: 10/15/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	166.10	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	30.64	0.00	0.00	1,908.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	210.74	0.00	0.00	1,908.69
Anticipated crop nutrient removal	64.50	10.95	69.75	0.00
Actual crop nutrient removal	171.39	29.01	185.90	0.00
Nutrient balance	39.34	-29.01	-185.90	1,908.69
Applied to removed ratio	1.23	0.00	0.00	0.00

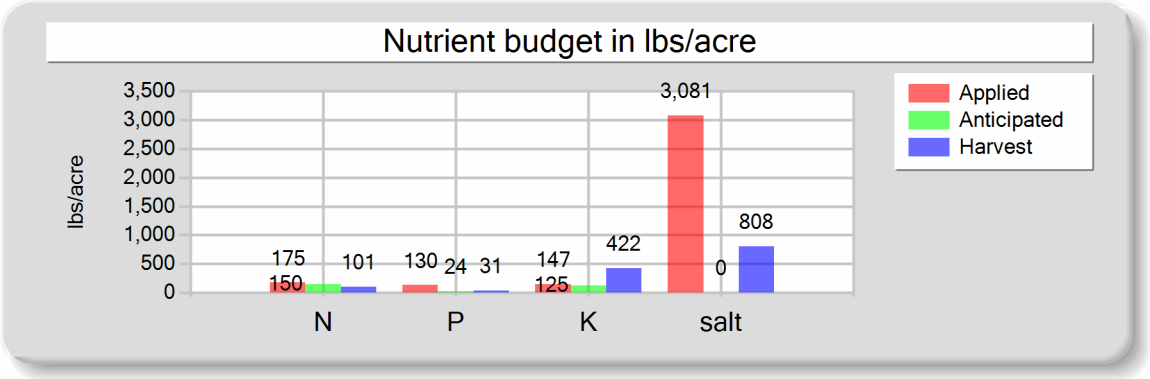
Fresh water applied	
45,744,400.00	*gallons*
1,684.61	*acre-inches*
44.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/31/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	168.12	129.94	147.36	3,081.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.12	129.94	147.36	3,081.13
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	101.02	31.11	421.96	808.41
Nutrient balance	74.10	98.84	-274.60	2,272.72
Applied to removed ratio	1.73	4.18	0.35	3.81

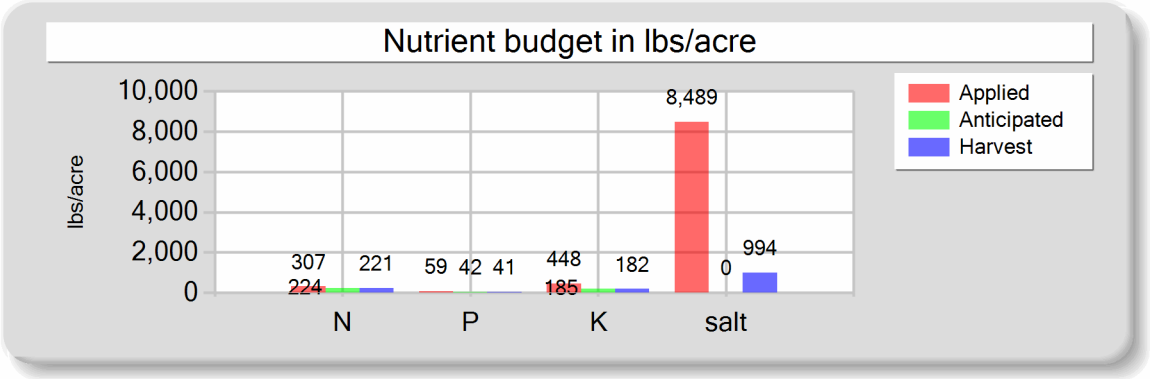
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 2 - 06/11/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	162.40	48.12	244.88	5,748.35
Process wastewater	107.70	11.12	202.74	1,362.38
Fresh water	21.89	0.00	0.00	1,378.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.99	59.24	447.63	8,488.90
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	220.80	41.40	182.16	993.60
Nutrient balance	86.19	17.84	265.47	7,495.30
Applied to removed ratio	1.39	1.43	2.46	8.54

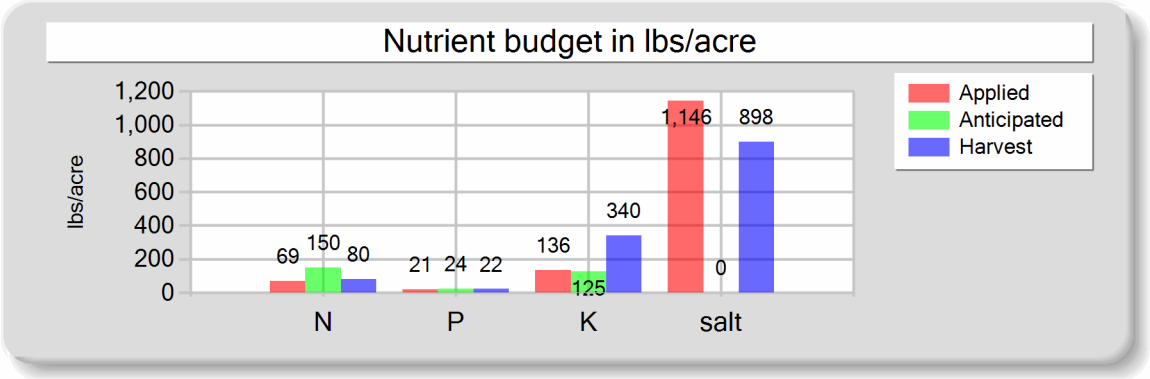
Fresh water applied	
46,068,000.00	*gallons*
1,696.53	*acre-inches*
32.01	*inches/acre*

Process wastewater applied	
1,710,000.00	*gallons*
62.97	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/31/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	61.54	20.78	135.87	1,145.56
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	68.54	20.78	135.87	1,145.56
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	79.96	21.54	340.30	898.12
Nutrient balance	-11.42	-0.76	-204.43	247.45
Applied to removed ratio	0.86	0.96	0.40	1.28

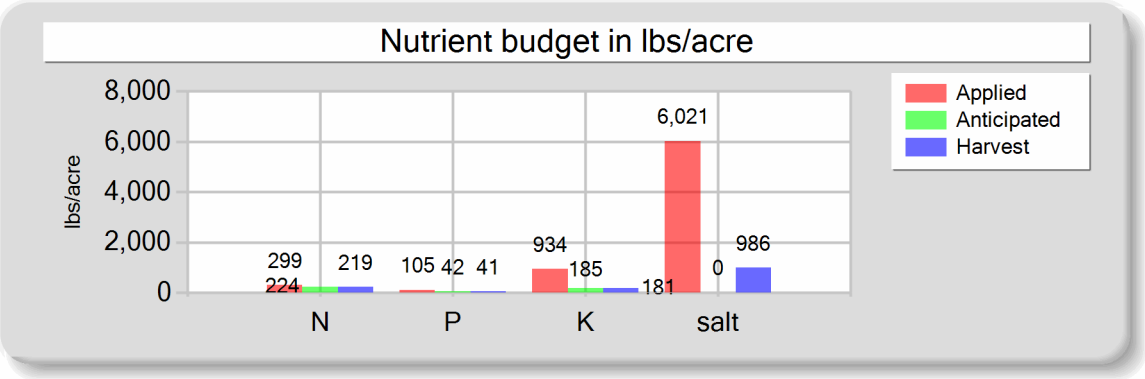
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
4,230,000.00	*gallons*
155.78	*acre-inches*
2.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/13/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	204.77	98.77	827.51	3,878.58
Process wastewater	56.68	5.85	106.71	717.04
Fresh water	22.65	0.00	0.00	1,425.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.10	104.62	934.22	6,021.18
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	219.20	41.10	180.84	986.40
Nutrient balance	79.90	63.52	753.38	5,034.78
Applied to removed ratio	1.36	2.55	5.17	6.10

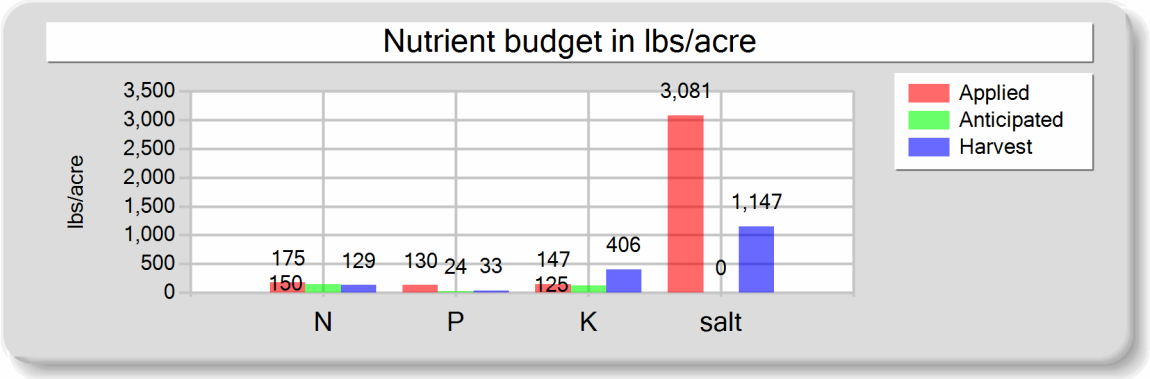
Fresh water applied	
47,652,000.00	*gallons*
1,754.86	*acre-inches*
33.11	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
0.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/01/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	168.12	129.94	147.36	3,081.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.12	129.94	147.36	3,081.13
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	129.44	33.49	405.72	1,146.89
Nutrient balance	45.68	96.46	-258.36	1,934.25
Applied to removed ratio	1.35	3.88	0.36	2.69

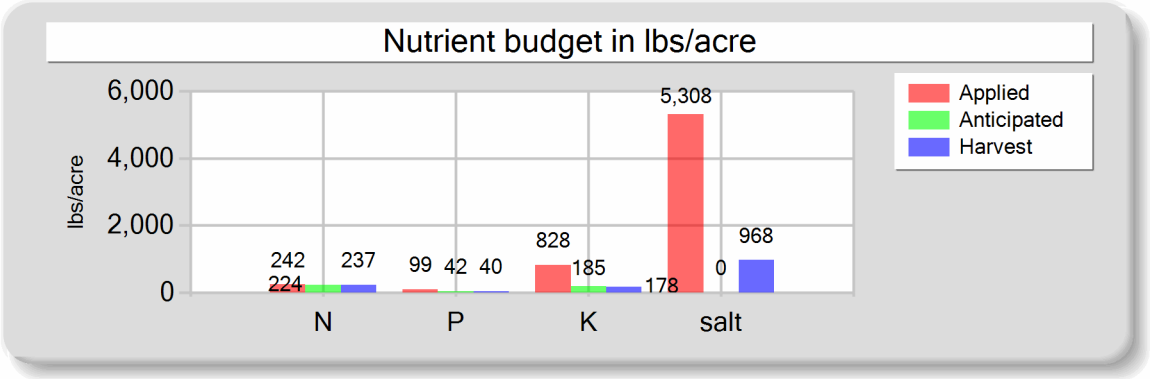
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 4 - 06/15/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	204.77	98.77	827.51	3,878.58
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.71	0.00	0.00	1,429.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.48	98.77	827.51	5,308.09
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	236.72	40.35	177.54	968.40
Nutrient balance	5.76	58.42	649.97	4,339.69
Applied to removed ratio	1.02	2.45	4.66	5.48

Fresh water applied	
47,784,000.00	*gallons*
1,759.72	*acre-inches*
33.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,100.00	19,400.00	22,000.00	54,900.00	18,800.00	5,290.00	8,700.00	3,440.00		46.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	8,200.00	68,700.00	14,400.00	8,300.00	7,300.00	4,600.00	4,340.00		32.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,900.00	5,600.00	28,500.00	11,800.00	6,300.00	2,900.00	4,600.00	1,790.00		66.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	4,840.00	41,300.00							17.20
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	92.40	2.80	0.00	0.00	31.20	204.00	27.90	26.10	39.70	660.00	0.00	65.20	137.00	1,920.00	1,720
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	588.00	19.60	0.00	0.00	42.60	250.00								1,060.00	900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	400.00	333.00	0.00	0.00	41.30	753.00								6,600.00	5,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	437.00	328.00	0.00	0.00	53.90	679.00								5,950.00	4,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW F1

IW F1

Sample description: IW F1

Sample date: 11/28/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.05	0.00	3.05								333.00	190
DL	0.50	0.50	0.50								1.00	10

IW Milk Barn

IW Milk Barn

IW Milk Barn

Sample description: IW Milk Barn

Sample date: 11/28/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.98	0.00	2.98								331.00	190
DL	0.50	0.50	0.50								1.00	101

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			75.00				
DL			4.00				

Field 2

Soil

Sample and source description: Soil

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			121.00				
DL			4.00				

Field 3

Field 3

Soil

Sample and source description: Soil

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 10/15/2017: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	65,000.00	11,000.00	70,500.00		0.00
DL	100.00	100.00	100.00		0.01

Field 2 - 10/31/2022: Oats, silage-soft dough

Field 2 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,280.00	1,010.00	13,700.00		6.73
DL	100.00	100.00	100.00		0.01

Field 2 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	750.00	3,300.00		6.00
DL	100.00	100.00	100.00		0.01

Field 3 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,190.00	590.00	9,320.00		9.11
DL	100.00	100.00	100.00		0.01

Field 3 - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	750.00	3,300.00		6.00
DL	100.00	100.00	100.00		0.01

Field 4 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,020.00	1,040.00	12,600.00		7.88
DL	100.00	100.00	100.00		0.01

Field 4 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	750.00	3,300.00		6.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

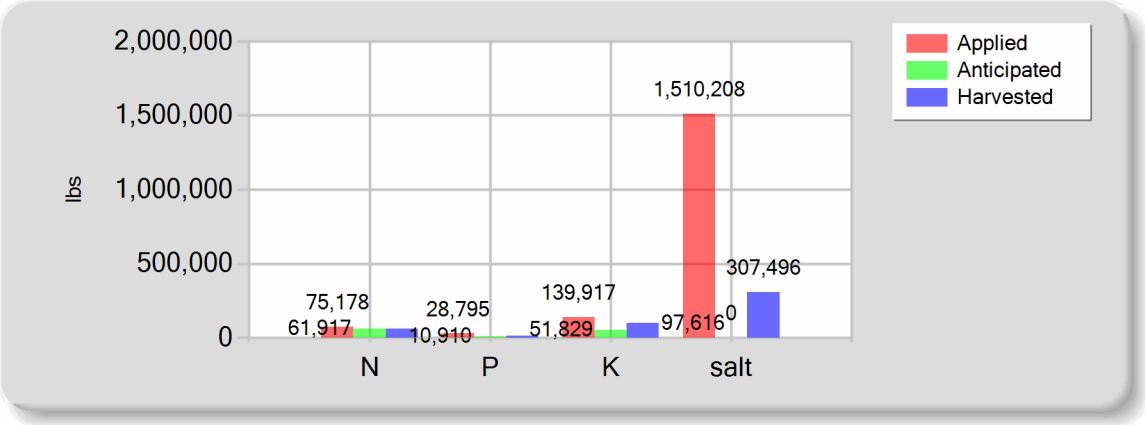
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

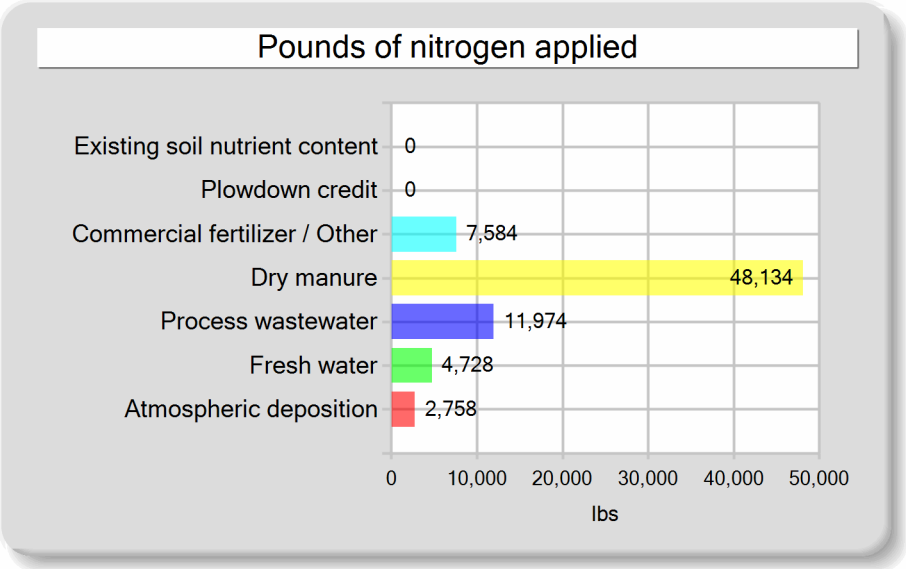
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

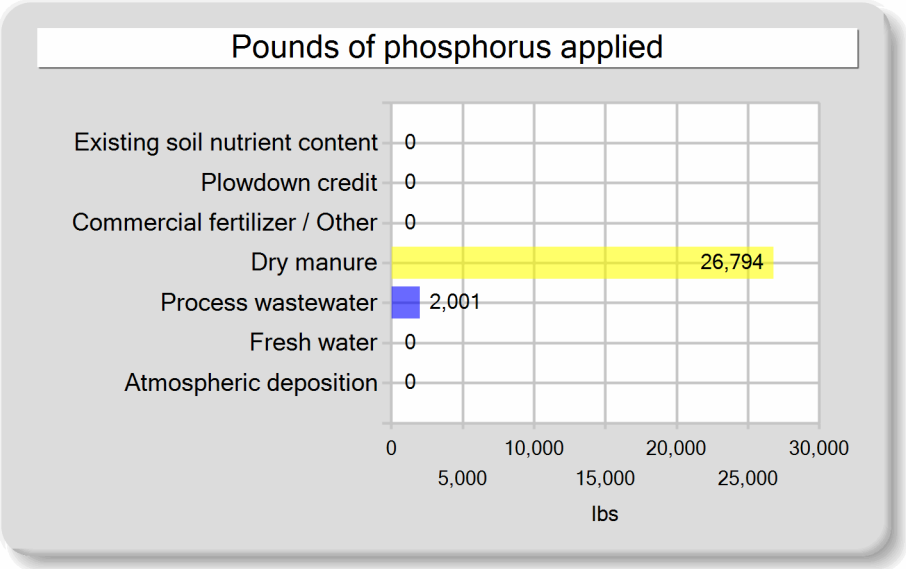
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	7,583.80	0.00	0.00	0.00
Dry manure	48,133.66	26,794.00	116,315.06	1,042,392.20
Process wastewater	11,973.84	2,000.87	23,601.75	170,923.96
Fresh water	4,728.32	0.00	0.00	296,891.70
Atmospheric deposition	2,758.00	0.00	0.00	0.00
Total nutrients applied	75,177.62	28,794.87	139,916.81	1,510,207.86
Anticipated crop nutrient removal	61,917.00	10,910.10	51,829.20	0.00
Actual crop nutrient removal	58,832.05	12,178.61	97,615.82	307,495.91
Nutrient balance	16,345.57	16,616.27	42,300.98	1,202,711.95
Applied to removed ratio	1.28	2.36	1.43	4.91

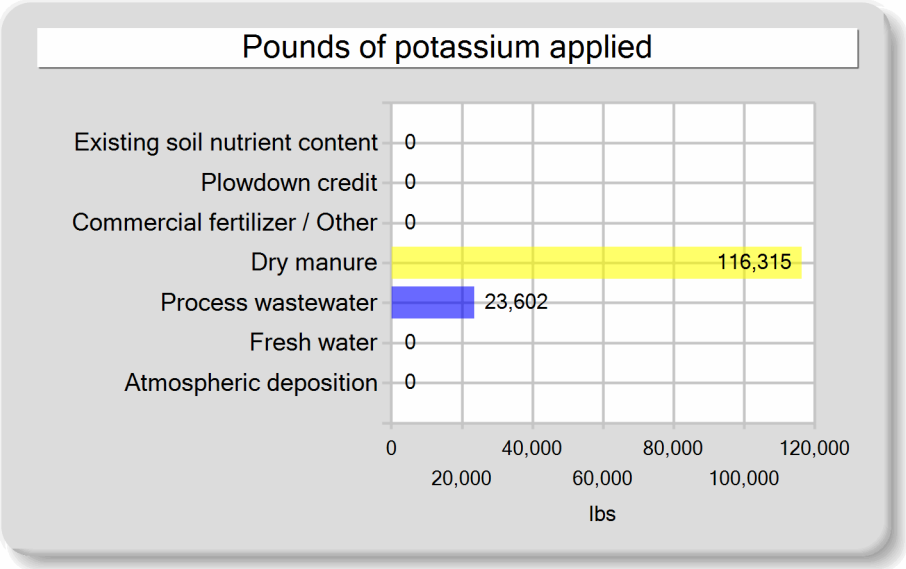
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

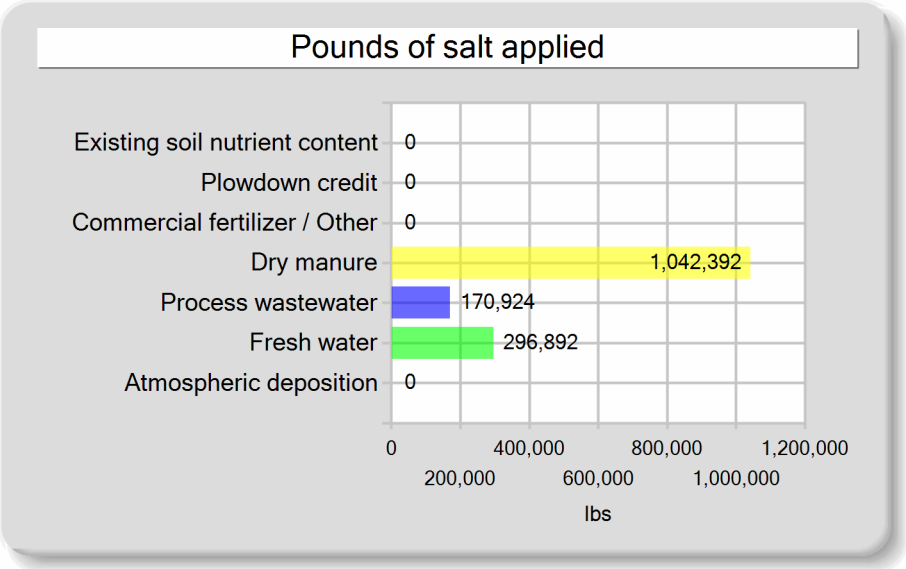


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 2 Oats (1.73) had a high nitrogen field ratio as field failed to meet anticipated nitrogen removal rate of 150 pounds of nitrogen/acre. The subject field removed only 101 pounds of nitrogen/acre. Yield for the field was adequate however a low nitrogen value from the plant tissue lab analysis resulted in only 6.56 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton which ultimately was the main reason for the high nitrogen field ratio. Nitrogen application to the subject field was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

Well samples were missed in 2023. Well samples used for this report were from 2022. Corn samples were missed in 2023, book value corn samples were used for this report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Tony Borba

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Tony Borba	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
5-28-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.